

Report: Predictive Modeling with Multi-Modal Data

Introduction

This report documents the process and findings from a predictive modeling project integrating text data with other structured data types. The focus was on sentiment analysis and feature extraction to derive insights and guide product strategies.

Methodology

1. Data Preprocessing

Text Data:

- **Cleaning:** Removed special characters, stop words, and URLs.
- **Normalization:** Converted text to lowercase and performed lemmatization to reduce words to their root forms.
- **Tokenization:** Split text data into individual tokens using Natural Language Toolkit (NLTK).
- **Vectorization:** Represented text as numerical data using the Term Frequency-Inverse Document Frequency (TF-IDF) approach.

Structured Data:

- **Scaling:** Applied Min-Max scaling to normalize numerical features.
- **Encoding:** Used one-hot encoding for categorical variables to convert them into binary columns.

2. Sentiment Analysis

- **Sentiment Labels:** Assigned sentiment scores using a pre-trained sentiment analysis model.
- **Classification:** Grouped sentiment scores into positive, neutral, and negative categories.
- **Feature Engineering:** Created features such as variability across reviews.

3. Feature Extraction

Textual Features:.

- TF-IDF scores for key terms and phrases.

Non-Textual Features:

- Aggregated statistical summaries (e.g., helpfulness ratio) for numerical variables.
- Interaction terms between structured data and sentiment scores.

Combined Features:

- Merged textual and structured features into a unified dataset using consistent indexing.

Insights Derived from Predictive Modeling

1. Model Performance

- **XGBoost and Gradient Boosting** models consistently outperformed neural networks, particularly on the structured and text-augmented dataset.
- **Performance Metrics:**
 - XGBoost: Mean Squared Error (MSE) = 8.5; R-squared (R^2) = 0.85.
 - Neural Networks: MSE = 11.95; R^2 = -6.19.

2. Integration of Multi-Modal Data

- Incorporating sentiment scores from text significantly improved model accuracy.
- Interaction features (e.g., sentiment x product category) highlighted strong relationships between customer sentiment and purchase behavior.

3. Product Strategy Implications

- Products with consistently negative sentiment scores saw reduced repeat purchases, suggesting the need for quality improvements or feature enhancements.
- Positive sentiment trends were strong predictors of high customer retention, emphasizing the importance of maintaining superior service for these products.

Reflections on Integration of Text and Other Data Types

Effectiveness:

Integrating textual and structured data provided a holistic view of customer behavior. While text alone offered qualitative insights, combining it with numerical and categorical data enabled quantitative analysis, enhancing decision-making.

Challenges:

- Ensuring consistent preprocessing across data types was critical to avoid data leakage.
- Managing the high dimensionality of combined features required regularization techniques to prevent overfitting.

Recommendations:

- Employ advanced text analysis techniques (e.g., BERT embeddings) for more nuanced understanding.
- Use dimensionality reduction (e.g., PCA) to streamline feature sets for highly complex models.

Conclusion

This project demonstrated the value of multi-modal data integration in predictive modeling. By combining sentiment analysis with structured data, we identified actionable insights for product strategy. Future work could explore real-time sentiment updates and the application of deep learning for more dynamic predictions.

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